

A Short Guide to Statistical Rethinking²

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A meta post introducing my solutions to the fantastic [excellent second edition](#) of “Statistical Rethinking” by [Richard McElreath](#), a.k.a. *Statistical Rethinking*². Also discusses strategies to keep up with the material, mostly meant for self-study groups.

Background

As [detailed previously](#), I recently was part of a course centered around Bayesian modeling for the Icelandic COVID-19 pandemic. The Bayesian mindset needs no introduction, and this post is completely inadequate to explain why anyone should be interested (that’s what the book is for!). That said, especially for self-paced study groups, it might help to have some structure.

Solutions

These are meant to be sample solutions, and everyone **should** solve these for themselves. Each solution contains the packages used, as well as a colophon in the later posts to ensure reproducibility. Essentially this consists of four posts:

Week I

Covers the first four chapters {1,2,3,4}

Week II

Covers the next three chapters {5,6,7}

Week III

Covers five chapters {9,11,12}

Week IV

The last five chapters {13,14}

More concisely:

Chapter	Solutions
1. The Golem of Prague	N/A
2. Small Worlds and Large Worlds	here
3. Sampling the Imaginary	here
4. Geocentric Models	here
5. The Many Variables & The Spurious Waffles	here
6. The Haunted DAG & The Causal Terror	here
7. Ulysses’ Compass	here
8. Conditional Manatees	N/A
9. Markov Chain Monte Carlo	here
10. Big Entropy and the Generalized Linear Model	N/A

Chapter	Solutions
11. God Spiked the Integers	here
12. Monsters and Mixtures	here
13. Models With Memory	here
14. Adventures in Covariance	here
15. Missing Data and Other Opportunities	TBA
16. Generalized Linear Madness	TBA
17. Horoscopes	N/A

Pacing

The solutions compiled here were from an accelerated 4-week course covering the Statistical Rethinking² in four weeks. The book is more traditionally used in a full-semester course, so that should be kept in mind as well.

Resources

These are highly opinionated and the following list is in no way complete.

Canonical Content

- [Richard's Website](#)
- The [YouTube 2019 playlist](#)
- [Richard on Twitter](#)
- The [Stan page](#)

Additional Content

- An [introduction to Frequentist and Bayesian statistics](#) from LLNL by Kristin Lennox
- A [simple COVID-19 model](#) for Iceland
- More [complete COVID-19 model](#) for Iceland
- [Convergence Diagnostics for MCMC](#)
- Betancourt's [Conceptual Introduction to HMC](#)

Follow-up Courses

- [Bayesian Data Analysis](#)

Conclusions

This has been a short meta post which is essentially meant to collect content posted with dates in the past. Though this is not exactly a complete reference for beginners, it might still help people.